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/* Load the dataset into SAS */

FILENAME REFFILE '/home/u64189948/EDData/SelctedDataFilter2New_MergAmb.csv';

PROC IMPORT DATAFILE=REFFILE
  DBMS=CSV
  OUT=WORK.IMPORT;
  GETNAMES=YES;
RUN;

ODS RTF FILE='/home/u64189948/EDData/ED_Los_Analysis_Report.rtf' STYLE=JOURNAL;

PROC CONTENTS DATA=WORK.IMPORT; RUN;

/* Display variable names */
PROC CONTENTS DATA=WORK.IMPORT;
RUN;

/* Create a new dataset selecting specific columns */
DATA ED_Subset;
  SET WORK.IMPORT;
  KEEP "LOS rounded"N "Arrival Shift (1st: 12 AM)"N Gender EDLevel DispoType
  PatientsPerMD_Mean_Rd PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd
  "Ward Change"N "Arrival Day"N ESILevel "Age Group"N FinalDiagnosis TransportArrive FirstWard;
RUN;

/* Display the contents of the new subset */
PROC CONTENTS DATA=ED_Subset;
RUN;

/* Histogram for LOS rounded */
PROC UNIVARIATE DATA=ED_Subset;
  HISTOGRAM "LOS rounded"N;
RUN;

/* Density Plot for "LOS rounded" */
PROC SGPLOT DATA=ED_Subset;
  DENSITY "LOS rounded"N / TYPE=KERNEL;
  HISTOGRAM "LOS rounded"N / TRANSPARENCY=0.5;
  TITLE "Density Plot of LOS Rounded";
RUN;

/* Analyze characteristics of observations where LOS rounded > 24 */
PROC FREQ DATA=ED_Subset;
  TABLES FinalDiagnosis / NOROW NOCOL NOPERCENT;
  WHERE "LOS rounded"N > 24;
  TITLE "Final Diagnosis Distribution for LOS Rounded > 24";
RUN;

/* many observation more than 24 are related to mental illness */

/* Frequency analysis for FirstWard excluding NA */
PROC FREQ DATA=ED_Subset;
  TABLES FirstWard;
  WHERE FirstWard NE "NA";
  TITLE "Frequency Distribution of FirstWard (Excluding NA)";
RUN;

/* Create a final dataset excluding specific FinalDiagnosis values */
DATA ED_Final;
  SET ED_Subset;
  WHERE NOT (FinalDiagnosis IN ("Depression", "hallucinations", "schizophrenia",
  "behavioral dyscontrol", "Suicidal ideation",
  "Severe depression", "Psychosis", "Auditory hallucinations",
  "Suicidal ideation with recent attempt of han"));
RUN;

/* Analyze characteristics of observations where LOS rounded > 24 */
PROC FREQ DATA=ED_Final;
  TABLES FinalDiagnosis / NOROW NOCOL NOPERCENT;
  WHERE "LOS rounded"N > 24;
  TITLE "Final Diagnosis Distribution for LOS Rounded > 24";
RUN;

/* Create a new dataset excluding observations with LOS rounded > 24 */
DATA ED_Final2;
  SET ED_Final;
  WHERE "LOS rounded"N <= 24;
RUN;

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/* Density Plot for "LOS rounded" */
PROC SGPLOT DATA=ED_Final2;
  DENSITY "LOS rounded" N / TYPE=KERNEL;
  HISTOGRAM "LOS rounded" N / TRANSPARENCY=0.5;
  TITLE "Density Plot of LOS Rounded";
RUN;

/* Correlation analysis between selected variables */
PROC CORR DATA=ED_Final2;
  VAR PatientsPerMD_Mean_Rd PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd;
  TITLE "Correlation Analysis of PatientsPerMD_Mean_Rd, PatientsPerNURSE_Mean_Rd, and PatientsPerPA_RESIDENT_Mean_Rd";
RUN;

/* Scatter plots to check linearity assumption */
PROC SGPLOT DATA=ED_Final2;
  SCATTER X=PatientsPerMD_Mean_Rd Y="LOS rounded" N;
  TITLE "Scatter Plot: PatientsPerMD_Mean_Rd vs LOS Rounded";
RUN;

PROC SGPLOT DATA=ED_Final2;
  SCATTER X=PatientsPerNURSE_Mean_Rd Y="LOS rounded" N;
  TITLE "Scatter Plot: PatientsPerNURSE_Mean_Rd vs LOS Rounded";
RUN;

PROC SGPLOT DATA=ED_Final2;
  SCATTER X=PatientsPerPA_RESIDENT_Mean_Rd Y="LOS rounded" N;
  TITLE "Scatter Plot: PatientsPerPA_RESIDENT_Mean_Rd vs LOS Rounded";
RUN;

/* Frequency analysis for Arrival Day */
PROC FREQ DATA=ED_Final2;
  TABLES "Arrival Day" N;
  TITLE "Frequency Distribution of Arrival Day";
RUN;

/* Exclude observations where Arrival Day is Sat or Sun */
DATA ED_Final3;
  SET ED_Final2;
  WHERE "Arrival Day" N NOT IN ("Sat", "Sun");
RUN;

/* Log Transformation to Improve Linearity */
DATA ED_Transformed;
  SET ED_Final3;
  Log_LOS_Rounded = LOG("LOS rounded" N + 1);
  Log_PatientsPerMD = LOG(PatientsPerMD_Mean_Rd + 1);
  Log_PatientsPerNURSE = LOG(PatientsPerNURSE_Mean_Rd + 1);
  Log_PatientsPerPA_RESIDENT = LOG(PatientsPerPA_RESIDENT_Mean_Rd + 1);
RUN;

/* Scatter plots for transformed variables */
PROC SGPLOT DATA=ED_Transformed;
  SCATTER X=Log_PatientsPerMD Y=Log_LOS_Rounded;
  TITLE "Scatter Plot: Log_PatientsPerMD vs Log_LOS_Rounded";
RUN;

PROC SGPLOT DATA=ED_Transformed;
  SCATTER X=Log_PatientsPerNURSE Y=Log_LOS_Rounded;
  TITLE "Scatter Plot: Log_PatientsPerNURSE vs Log_LOS_Rounded";
RUN;

PROC SGPLOT DATA=ED_Transformed;
  SCATTER X=Log_PatientsPerPA_RESIDENT Y=Log_LOS_Rounded;
  TITLE "Scatter Plot: Log_PatientsPerPA_RESIDENT vs Log_LOS_Rounded";
RUN;

PROC CONTENTS DATA=ED_Transformed; RUN;

/* Regression Analysis with correctly formatted variables */
PROC GLM DATA=ED_Transformed;
  CLASS "Age Group" N Gender ESILevel "Arrival Shift (1st: 12 AM)" N "Arrival Day" N DispoType FirstWard "Ward Chang
  MODEL Log_LOS_Rounded = "Age Group" N Gender ESILevel Log_PatientsPerMD
    Log_PatientsPerNURSE Log_PatientsPerPA_RESIDENT
    "Arrival Shift (1st: 12 AM)" N "Arrival Day" N DispoType
    FirstWard "Ward Change" N EDLevel TransportArrive;
  TITLE "Regression Analysis of Log_LOS_Rounded with Categorical Variables";
RUN;

/* Machine learnign Work*/
/* Convert LOS rounded to categorical in ED_Final3. If greated than 4 =1 */

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DATA ED_Final3_Cat;
  SET ED_Final3;
  IF "LOS rounded"N > 4 THEN LOS_Cat = 1;
  ELSE LOS_Cat = 0;
RUN;

/* Display first few observations of ED_Final3_Cat */
PROC PRINT DATA=ED_Final3_Cat (OBS=10);
  TITLE "First 10 Observations of ED_Final3_Cat with LOS Categorized";
RUN;

/* Split ED_Final3_Cat into Training (70%) and Testing (30%) */
PROC SORT DATA=ED_Final3_Cat;
  BY RandNum;
RUN;

DATA Train Test;
  SET ED_Final3_Cat NOBS=TotalObs;
  IF _N_ <= 0.7 * TotalObs THEN OUTPUT Train;
  ELSE OUTPUT Test;
RUN;

/* Display first few observations of training and testing datasets */
PROC PRINT DATA=Train (OBS=10);
  TITLE "First 10 Observations of Training Dataset";
RUN;

PROC PRINT DATA=Test (OBS=10);
  TITLE "First 10 Observations of Testing Dataset";
RUN;

/* Logistic Regression on Training Data */
PROC LOGISTIC DATA=Train DESCENDING;
  CLASS "Age Group"N Gender ESILevel "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive / PARAM=REF;
  MODEL LOS_Cat = "Age Group"N Gender ESILevel PatientsPerMD_Mean_Rd
    PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd
    "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive / EXPB;
  OUTPUT OUT=Train_Predicted P=Pred_Prob;
  TITLE "Logistic Regression on Training Data";
RUN;

/* Logistic Regression on Testing Data */
PROC LOGISTIC DATA=Test DESCENDING;
  CLASS "Age Group"N Gender ESILevel "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive / PARAM=REF;
  MODEL LOS_Cat = "Age Group"N Gender ESILevel ESILevel PatientsPerMD_Mean_Rd
    PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd
    "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive / EXPB;
  OUTPUT OUT=Test_Predicted P=Pred_Prob;
  TITLE "Logistic Regression on Testing Data";
RUN;

/* ROC Curve Analysis */
PROC LOGISTIC DATA=Test DESCENDING PLOTS=ROC;
  MODEL LOS_Cat = "Age Group"N Gender ESILevel Log_PatientsPerMD
    Log_PatientsPerNURSE Log_PatientsPerPA_RESIDENT
    "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive;
  TITLE "ROC Curve Analysis for Logistic Regression";
RUN;

/* XGBoost Model */

PROC HPSPLIT DATA=Train SEED=1234;
  CLASS "Age Group"N Gender ESILevel "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive;
  MODEL LOS_Cat = PatientsPerMD_Mean_Rd
    PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd
    "Arrival Shift (1st: 12 AM)"N "Arrival Day"N DispoType
    FirstWard "Ward Change"N EDLevel TransportArrive;
  PRUNE COSTCOMPLEXITY;
  TITLE "Boosted Trees (Alternative to XGBoost) for LOS_Cat Prediction";
RUN;

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PROC HPSPLIT DATA=Test SEED=1234;
  CLASS "Age Group" N Gender ESILevel "Arrival Shift (1st: 12 AM)" N "Arrival Day" N DispoType
    FirstWard "Ward Change" N EDLevel TransportArrive;
  MODEL LOS_Cat = PatientsPerMD_Mean_Rd
    PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd
    "Arrival Shift (1st: 12 AM)" N "Arrival Day" N DispoType
    FirstWard "Ward Change" N EDLevel TransportArrive;
  PRUNE COSTCOMPLEXITY;
  TITLE "Boosted Trees (Alternative to XGBoost) for LOS_Cat Prediction";
RUN;

/* Check accurarcy of the model */

/* Step 1: Create a binary classification table for Logistic Regression */
PROC FORMAT;
  VALUE pred_fmt 0 = "No Prolonged LOS"
    1 = "Prolonged LOS";
RUN;

DATA Test_Predicted_Eval;
  SET Test_Predicted;
  Predicted_Class = (Pred_Prob >= 0.5); /* Threshold at 0.5 */
  FORMAT LOS_Cat Predicted_Class pred_fmt.;
RUN;

/* Confusion Matrix and Performance for Logistic Regression */
PROC FREQ DATA=Test_Predicted_Eval;
  TABLES LOS_Cat * Predicted_Class / NOCOL NOROW NOPERCENT CHISQ;
  TITLE "Confusion Matrix for Logistic Regression";
RUN;

/* Compute Accuracy for Logistic Regression */
PROC MEANS DATA=Test_Predicted_Eval N MEAN;
  VAR LOS_Cat;
  TITLE "Accuracy of Logistic Regression Model";
RUN;

/* ROC Curve and AUC Calculation for Logistic Regression */
/* Compute ROC and AUC for Logistic Regression */
PROC LOGISTIC DATA=Test_Predicted_Eval DESCENDING PLOTS(ONLY)=ROC;
  MODEL LOS_Cat = Pred_Prob / OUTROC=ROC_Logistic;
  TITLE "ROC Curve Analysis for Logistic Regression";
RUN;

/* Evaluate XGBoost Predictions*/

/* Apply Boosted Trees (XGBoost Alternative) to Test Data */
/* Apply the trained XGBoost (HPSPLIT) Model to the Test Data */
PROC HPSPLIT DATA=Test SEED=1234;
  CLASS "Age Group" N Gender ESILevel "Arrival Shift (1st: 12 AM)" N "Arrival Day" N DispoType
    FirstWard "Ward Change" N EDLevel TransportArrive;
  MODEL LOS_Cat = PatientsPerMD_Mean_Rd
    PatientsPerNURSE_Mean_Rd PatientsPerPA_RESIDENT_Mean_Rd
    "Arrival Shift (1st: 12 AM)" N "Arrival Day" N DispoType
    FirstWard "Ward Change" N EDLevel TransportArrive;
  PRUNE COSTCOMPLEXITY;
  SCORE OUT=Test_XGBoost_Pred;
  TITLE "Apply Boosted Trees (XGBoost Alternative) on Test Data";
RUN;

/* Convert Predicted Probabilities to Binary Predictions */
DATA Test_XGBoost_Eval;
  SET Test_XGBoost_Pred;
  Predicted_Class = (P_LOS_Cat1 >= 0.5); /* Assuming P_LOS_Cat1 is the probability for LOS_Cat=1 */
RUN;

PROC FREQ DATA=Test_XGBoost_Eval;
  TABLES LOS_Cat * Predicted_Class / NOCOL NOROW NOPERCENT CHISQ;
  TITLE "Confusion Matrix for XGBoost Model";
RUN;

ODS RTF CLOSE; /* If using Word */

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